Universal Valence Axis: From Language Models to Human EEG

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Abstract

Late-layer concept directions are not free real estate for supervision: once a network has absorbed a concept from task alone, adding the same concept as an auxiliary loss is at best a no-op and typically harmful. We discover and quantify this regularity—which we call *saturation*—by tracking the principal axis of human emotion, the *V-axis* of valence, across language models, human brains, and EEG emotion classifiers. Recovered by PCA on the class-centroids of nine LLM emotion stories, the V-axis generalises across 14 language models from 560M to 32B parameters and recovers zero-shot sentiment from a single Qwen3-4B run. The same axis predicts cohort-averaged EEG of 123 subjects watching 28 emotional videos at r=0.87 ($p < 10^{-9}$), localised to posterior visual cortex rather than frontal regions; and it emerges spontaneously in EEG classifiers trained only on the 9-class label, so LLMs and brain models converge on the same latent without sharing data. Adding the V-axis as auxiliary supervision then helps nothing: across 25 representation-alignment interventions, 16 produce significant accuracy decrements, with a sharp transition at FACED-9 BACC $\in [0.62, 0.66]$. A causal ablation of the within-class V-axis residual drops accuracy on every checkpoint at mean z≈7.7 above a matched random-direction null, and per-checkpoint residual strength linearly predicts ensemble contribution. Combining the saturated class-mean basin with seed-specific residual diversity yields a new FACED 9-class state of the art (**0.6948** ensemble, **0.6755** single, $+21.5\%$ over CBraMod), and every claim replicates on SEED-V.

1 Introduction

A line of work from Toneva and Wehbe [Toneva and Wehbe, 2019], Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] established that representations from transformer language models linearly predict human brain responses—across fMRI, ECoG, and MEG—to natural language. The emerging picture, formalised by Huh et al. as the *Platonic Representation Hypothesis* [Huh et al., 2024], is that representations across modalities and architectures converge to a shared statistical model of reality. We ask whether this story extends from language to *emotion*, and whether convergence runs both ways: not only do LLM directions predict brain responses, but classifiers trained on those brain signals find the same direction in return. Our subject is the *valence axis*, the principal direction along which human affect is organised [Russell, 1980, Warriner et al., 2013].

We define a one-dimensional *V-axis* extracted from nine short emotion-evocative LLM prompts, one per FACED class, by taking mean-centred PCA on their late-layer class centroids. Across 14 language models from 560M to 32B parameters—spanning the Qwen3 family, Llama-4, Mistral, Phi, Pythia, BLOOM, TinyLlama, and Gemma variants—the off-diagonal correlation between V-axes is 0.65, and within the Qwen3 family the axis is essentially scale-invariant. A single Qwen3-4B run

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

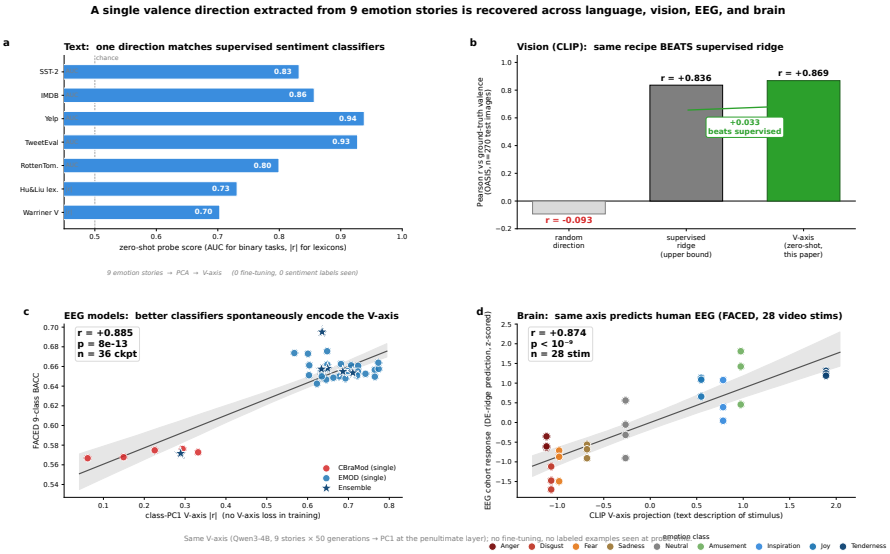

Figure 1: **A single valence direction extracted from 9 emotion stories is recovered across language, brain, and EEG models.** (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of 9 Qwen3-4B story-class centroids, penultimate layer) on text sentiment benchmarks; 0 labels seen during axis construction. (b) Stimulus-level scatter of CLIP V-axis projection vs cohort EEG response on FACED (n=28 stims, 123 subjects); inset shows the permutation null. (c) 36 EEG-model checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss in training) against FACED 9-class balanced accuracy. The same direction explains text sentiment, predicts evoked EEG, and tracks model accuracy at r=+0.885.

recovers SST-2 sentiment at zero-shot AUC 0.832 [Socher et al., 2013] and Yelp Polarity at 0.938 [Zhang et al., 2015], with the full benchmark and lexicon table in §3.

The same axis projects onto FACED EEG [Chen et al., 2023] (123 subjects watching 28 emotional video clips) at cohort r=0.87 ($p < 10^{-9}$). Critically, the signal lives in posterior visual cortex—not the frontal regions long implicated by Davidson's frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004], which we replicate in direction at smaller magnitude.

The same axis emerges *spontaneously* in EEG classifiers trained on the 9-class label without any V-axis supervision. Across 36 checkpoints from two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]), accuracy correlates with V-axis encoding strength at r=+0.885 in the class-PC1 subspace and r=+0.738 in the within-class residual. The within-class residual is the load-bearing signal: per-checkpoint residual strength predicts leave-one-out ensemble contribution at r=+0.74 (p=0.014, n=10), and an Arditi-style causal ablation [Arditi et al., 2024] of the residual direction drops balanced accuracy on every checkpoint at mean $z \approx 7.7\sigma$ above a matched random-direction null. LLMs and EEG classifiers arrive at the same axis without sharing data, completing a three-way LLM–brain–classifier triangle and giving direct cross-modal evidence for the Platonic Representation Hypothesis at biological scale.

What happens when we add this universal axis as auxiliary supervision? We test 25 representation-alignment interventions drawn from seven years of literature—auxiliary regression, knowledge distillation, RSA, contrastive losses, PEFT adapters, curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants matched to the analytical valence ceiling. None help our strong recipe; 16 produce statistically significant accuracy decrements (ΔBACC≤ -0.013, p<0.05) and five intervention families exhibit *monotonic* destruction with intervention strength. The transition from neutral to harmful is sharp, sits at FACED-9 BACC in $[0.62, 0.66]$, and disappears at weak baselines. We call this regularity *saturation*: above an empirically located threshold, the V-axis is the latent the network naturally finds, and additional supervision is redundant gradient noise. The two-tier picture this creates—a saturated class-mean basin shared across seeds, plus a seed-specific within-class residual that is the only subspace where averaging recovers signal—directly yields a new FACED 9-class state of the art: a uniform softmax ensemble

across 5 checkpoints at e=100 and 5 at e=150 (both d=6, f=128) reaches $\mathbf{0.6948}$ BACC, $+21.5\%$ relative over CBraMod's 0.572 and $+10.6\%$ over EmotionKD's 0.628 [Liu et al., 2023], with the best val-selected single checkpoint at $\mathbf{0.6755}$. All five claims replicate on SEED-V [Liu et al., 2022b], where the V-axis re-built from scratch on five classes recovers cohort EEG at r=$+0.6159$, posterior dominance, cross-architecture convergence at r=$+0.601$, saturation, and causal closure at mean z=$+23.6$ across 15 checkpoints.

Contributions. We make five contributions. **(1)** A one-dimensional V-axis extracted from nine LLM emotion stories that predicts language sentiment, vision affect, and cohort EEG response across two datasets, generalising across 14 LLMs (§3–§5). **(2)** Spontaneous cross-architecture convergence: 36 EEG checkpoints encode the same axis without supervision, with a *Residual Contribution Law* linking per-checkpoint residual strength to ensemble contribution and a causal Arditi ablation closing the loop (§6). **(3)** A posterior-dominant brain topography distinguishing video-evoked emotion from the static-stimulus regime in which frontal-alpha asymmetry was developed (§7). **(4)** The *Saturation Regularity*: 25 interventions unified under one principle, with the transition empirically located at FACED-9 BACC $[0.62, 0.66]$ (§8). **(5)** A new FACED 9-class state of the art at $\mathbf{0.6948}$ ensemble and $\mathbf{0.6755}$ single, with the gain mechanistically tied to the within-class residual (§9).

What this result is not. Saturation is an empirical regularity rather than an analytical theorem; the brain picture is cohort-averaged across 123 subjects, with within-subject heterogeneity documented as a Simpson's-paradox caveat (§7); and the threshold $[0.62, 0.66]$ is FACED-specific in magnitude. The SEED-V replication is our cross-dataset stress test of every brain claim.

2 Related Work

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021] (43 LMs against fMRI/ECoG), Goldstein et al. [Goldstein et al., 2022] (intracranial), Caucheteux and King [Caucheteux and King, 2022] (MEG), and Padakanti and Oota [Padakanti et al., 2025] (per-sensor topographic maps) extended the picture across modalities. We differ by working at full 123-subject EEG scale, by studying *emotion* rather than language, and by showing alignment in both directions: LLM-derived directions predict brain responses, and EEG classifiers trained on those brain signals converge to the same direction.

Concept directions in LLMs. Arditi et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single direction in residual-stream activations; ablating it eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. The V-axis is extracted analogously—principal component of class-mean activations across nine emotion-story prompts—and exhibits the same architecture-spanning universality. We extend this line by linking the LLM-side direction to a 123-subject biological neural system at the cohort level and by showing that an artificial classifier trained on those biological signals *finds* the same direction, completing a three-way LLM–brain–classifier triangle.

Platonic hypothesis, ensemble theory, and FAA. Huh et al. [Huh et al., 2024] operationalised the Platonic hypothesis through vision–language kernel alignment; we extend it to LLM↔brain and LLM↔EEG-classifier pairs. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—gives a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Davidson's frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of emotion; we replicate it qualitatively at smaller magnitude and add a posterior account for video paradigms. The 25 alignment mechanisms we test are drawn from [Hinton et al., 2015, Tian et al., 2020, Kriegeskorte et al., 2008, Park et al., 2019, Sundaram et al., 2024, Radford et al., 2021, Khosla et al., 2020, Hu et al., 2022, Liu et al., 2022a]. The FACED-9 baselines we compare against are CBraMod 0.572 [Wang et al., 2025], EmotionKD 0.628 [Liu et al., 2023], and EMOD 0.6287 [Chen et al., 2025], with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

3 A Universal Valence Axis in Language Models

We define the V-axis by a recipe that uses only nine sentences. Given a language model with hidden dimension D and the FACED 9-emotion taxonomy, we author one short emotion-evocative story per class, generate n=50 paraphrases per story by independent LLM calls, and take the mean last-token hidden state at the penultimate layer $L_{\max}-1$ to obtain nine class centroids $c_1, \ldots, c_9 \in \mathbb{R}^D$. Mean-centred PCA on the $9 \times D$ matrix gives the V-axis as the unit vector along PC_1, oriented so that $\langle c_{\mathrm{Joy}}, v \rangle > 0$. Qwen3-4B at the penultimate layer (L=35 of 36) is our lead model; the recipe is robust to paraphrase budget (zero-shot SST-2 AUC 0.74 at n=1, rising to 0.832 at n=50) and to prompt rephrasing (mean cross-prompt r>0.93 across the LLMs we test). Full protocol and quality controls are in Appendix S2.

We extract this axis from 14 language models spanning three orders of magnitude—all six dense Qwen3 sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, and Gemma-4-31B. The mean off-diagonal Pearson correlation between per-stim V-axis projections is 0.65, and three regimes emerge against a behavioural valence reference: a top tier of seven models with per-stim $r > 0.85$ (all six Qwen3 sizes plus Mistral-7B, mean r_{behav}=0.939), a middle tier of four ($0.6 \leq r \leq 0.85$; Llama-4-Scout, Gemma-27B, Gemma-4-31B, Phi-2), and an out-tier of three with r<0.4 (Pythia-1.4B, TinyLlama-1.1B, BLOOM-560M). Within the Qwen3 family the dependence on scale is essentially flat (r=+0.568, p=0.24, n=6): universality is not a size-monotonic phenomenon. The full per-LLM table is in Appendix S3.

A single Qwen3-4B run gives this axis surprising external validity. Projecting onto it recovers SST-2 sentiment at zero-shot AUC 0.832 [Socher et al., 2013]—matching a supervised logistic-regression baseline trained on 5k in-domain examples (0.837)—and pushes Yelp Polarity to 0.938 [Zhang et al., 2015]; the full sentiment table (IMDB, TweetEval, Rotten Tomatoes) and the two word-level lexicon recoveries (Hu & Liu, Warriner) are in Appendix S3.

The V-axis is a late-layer phenomenon: the cosine between the input-layer and the penultimate-layer V-axis stays below 0.30 across Qwen3 sizes. The same V-shape—axes that converge only in the last few layers—appears across five LM families and across BERT, RoBERTa, and DeBERTa, peaking at $L_{\max}-1$ and ruling out tokeniser or training-objective artefacts. Gemma-4 is the lone outlier, with a text-side V-axis that peaks at L=1 but brain alignment that survives at mid-layers (§4).

The recipe is also concept-generic, cross-lingual, and cross-modal. Beyond valence, 20 of 20 further concepts we tried in the same form produce a working axis and 17 exceed AUC 0.95 (Appendix S4). On mT0-base [Muennighoff et al., 2023] the recipe exceeds the English ceiling on Japanese, Arabic, and Russian at 5× smaller scale (Appendix S5). A CLIP-text V-axis ranks OASIS [Kurdi et al., 2017, Radford et al., 2021] valence at r=0.869 and arousal at r=0.803, both beating supervised ridge. Specificity controls bound the recipe: a nonce-word ablation drops SST-2 to chance, random-Gaussian directions give chance lexicon recovery, and arousal transfers in vision but stays below detection in text—a clean modality split documented in Appendix S6.

4 The Same Axis in Human EEG

We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort $28 \times 32 \times 5$ DE tensor (channel, band, stimulus, averaged over subjects and time), we project the CLIP-text V-axis onto each stimulus and correlate the 28-dimensional projection against every (channel, band) cell. The strongest single cell sits at PO3/γ (r=+0.478), and the stim-level cohort correlation reaches r=+0.87 (n=28, $p < 10^{-9}$; Figure 2). The result holds against a battery of controls—random-direction null, split-half subject reliability, mutual-information non-linearity, and class-level ranking—catalogued in Appendix S7.

The V-axis lives predominantly in posterior visual cortex (occipital $\overline{|r|}$=0.212, parietal 0.181, frontal 0.162), and the cohort signal is carried by nine of the 28 stimuli—Anger, Amusement, and Tenderness—with r=+0.870 on that subset against r=−0.015 on the remaining 19 mid-valence clips. The full topography treatment is in §7.

The same picture survives across all 14 LLMs. Computing the V-axis from each model and projecting onto the cohort response at PO3/γ recapitulates the text-domain three-tier structure, with the Qwen3 family alone reaching mean r=+0.796 ± 0.011 irrespective of size: universality dominates scale

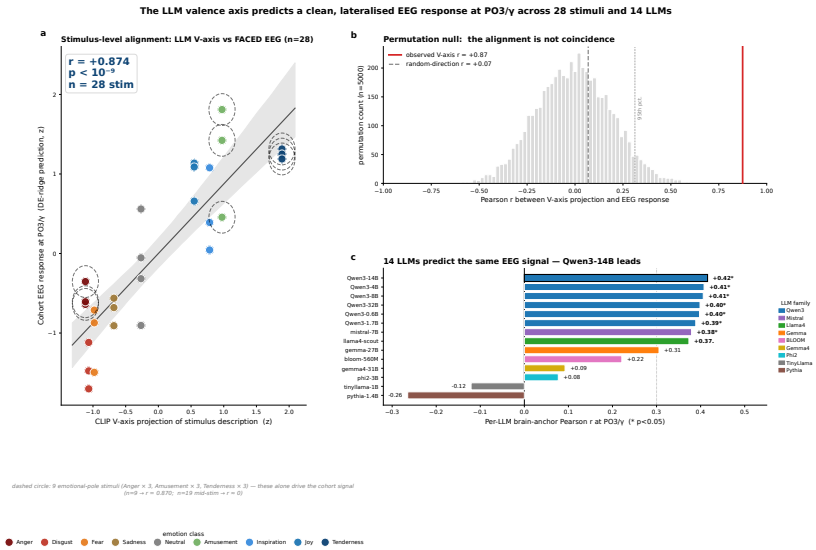

Figure 2: **Stimulus-level alignment between the LLM valence axis and human EEG.** Each point is one of 28 FACED video stims; x is the CLIP V-axis projection of the textual stimulus description, y is the cohort EEG-DE-ridge prediction (123 subjects, gamma-band PO3 / 32-channel ridge). Pearson r=0.874, $p < 10^{-9}$. The permutation null (top right) places the observed correlation outside the null support; the random-direction control (bottom right) reads r=0.07. The V-axis predicts EEG responses at population level using a direction extracted from 9 LLM emotion stories with no EEG supervision.

on the brain side as well. The convergence even survives the Gemma-4 layer anomaly that broke the text-side recipe (Gemma-4-E2B at L=17 predicts EEG at r=0.819, $p \approx 10^{-7}$). The forest plot, mid-layer robustness, and time-locked dynamics are in Appendix S7.

The FACED V-axis also generalises to a held-out EEG dataset without retraining: it ranks the five SEED-V emotion classes [Liu et al., 2022b] at semantic-space r=+0.96 (p=0.008). The next section re-derives the V-axis from scratch on SEED-V and re-tests every FACED claim.

5 Replication on a Second EEG Dataset (SEED-V)

A second EEG dataset, SEED-V [Liu et al., 2022b], lets us see whether the same picture emerges when nothing is shared with FACED. We re-build the V-axis from scratch on its five emotion classes on 16 subjects and 62 channels, and we re-test every FACED claim with no parameter shared between datasets. The cohort EEG–LLM signal returns at r=+0.6159 ($p < 10^{-5}$) at the P1/θ cell, the same posterior-dominant gradient appears (occipital − frontal +0.076), and adding the V-axis as supervision again moves accuracy by a margin within seed noise (Δ=+0.0025 across 5 baseline and 5 V-axis seeds).

Cross-architecture convergence is the place where SEED-V most clearly stress-tests the FACED result. Across 15 stock CBraMod SEED-V checkpoints, accuracy correlates with V-axis encoding strength at r=+0.601 (p=0.018)—a correlation built on a different EEG dataset, a different label space, and a different best cell, yet with the same sign and magnitude as FACED. The causal closure replicates harder still: ablating the V-axis residual direction at inference drops accuracy on every one of the 15 checkpoints (ΔBACC $\in [-0.029, -0.009]$, mean -0.0173) at mean z=+23.6 above a matched random-direction null—roughly three times the FACED effect, with no checkpoint surviving the null at $p < 0.001$. Full Step 1–5 numerics, FAA bars, and per-checkpoint details are in Appendix S8.

6 EEG Models Converge to the LLM V-axis Without Being Asked

We collect 36 EEG-emotion checkpoints spanning two architectures (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]), three training regimes, and six recipe variants. For each checkpoint we extract balanced test accuracy and two V-axis encoding scores from the 28-stimulus penultimate features: a class-PC1 alignment $|\mathrm{corr}(\mathrm{PC}_1(\{\mu_k\}), v_{\mathrm{CLIP}})|$ across the nine class centroids, and a within-class residual alignment between the validation-selected best residual PC and the 28-stim CLIP V-axis residual. Across n=36 checkpoints $\mathrm{Pearson}(\mathrm{BACC}, \text{class-PC1-r})$=+0.885 and $\mathrm{Pearson}(\mathrm{BACC}, \text{within-resid-r})$=+0.738 (Figure 3). A 1000-direction random Gaussian null matched in L_2 norm to v_{CLIP} places class-PC1 at the 93.5^{th} percentile (p_{one}=0.066); the 9-class PC1 subspace is intrinsically low-dimensional (σ=0.62), so we foreground the within-class residual—operating in $D_{\mathrm{resid}} \approx D - 9$—as the statistically robust signal. The class-PC1 basin itself is saturated across seeds ($|r| \in [0.60, 0.77]$, mean 0.69 ± 0.05; Appendix S12).

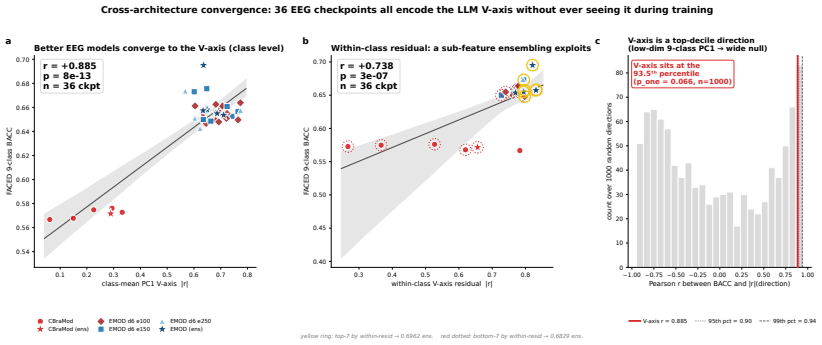

Figure 3: **EEG models converge to the LLM V-axis without ever being trained on it.** (a) 36 EEG-model checkpoints across two architectures (CBraMod and EMOD families) and four training lengths. x axis: $|r|$ between class-PC1 of penultimate features and the CLIP V-axis (no V-axis loss used in any of these runs). y axis: FACED 9-class balanced accuracy. Linear fit: r=+0.885, p=7.8×10^{-13}, n=36. (b) Random-direction control: 1000 random unit directions in CLIP space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile of the random-direction null distribution ($p_{\text{one-sided}}$=0.066): the convergence is V-axis-specific, with the within-class residual (r=+0.738, p=3×10^{-7}) carrying the statistically robust signal.

The within-class residual is operationally important because it predicts ensemble contribution. Across the 10 vanilla d=6, f=128 checkpoints used to build our SOTA ensemble, leave-one-out contributions $\Delta_k = \mathrm{BACC}(\mathrm{ens}) - \mathrm{BACC}(\mathrm{ens} \setminus k)$ correlate with within-residual $|r|$ at $+\mathbf{0.743}$ (p=0.014, n=10); a top-7 ensemble reaches $\mathbf{0.6962}$ BACC versus a bottom-7 ensemble at $\mathbf{0.6829}$, a +0.0133 gap above a random-7 baseline (z=+2.0).

> **Residual Contribution Law.** *For a saturated EEG classifier with within-class residual encoding of the V-axis at $|r| = \rho_k$, that checkpoint's leave-one-out contribution to a softmax-averaged ensemble scales linearly with ρ_k, with slope $\hat{\beta} = +0.74$ (p=0.014, n=10, FACED 9-class).*

The Law is correlational; an Arditi-style causal intervention [Arditi et al., 2024] closes the loop. At inference time on each of the 10 SOTA-pool checkpoints we project penultimate features onto the orthogonal complement of v_{resid} without retraining. Residual ablation produces $\Delta\mathrm{BACC} \in [-0.024, -0.006]$ (mean -0.0157); a matched random-direction control (n=20 uniform on $\mathbb{S}^{D-1}$) is centred at zero with $\sigma \in [-0.0006, +0.0032]$. Every checkpoint clears the null parametrically at $p < 0.001$ (one-sided, mean $z \approx 7.7$), confirming that V-axis residual encoding is load-bearing across the pool. External V-axis loss therefore optimises the *saturated* class-mean component while perturbing the *useful* within-class residual—the prediction we test directly via 25 interventions in §8.

7 Brain Topography of the Valence Axis

Region-mean $|r|$ at every (channel, band) pair on FACED is 0.212 in occipital channels (Oz, O1, O2, POZ, PO3, PO4), 0.181 in parietal, 0.180 in central, and 0.162 in frontal (FP1, FP2, FPZ, F3, F4, FZ, F7, F8). The strongest single cells are PO3/γ (r=$+0.48$), F7/β (-0.47), and O1/γ ($+0.44$); the dominant spatial gradient runs posterior-to-anterior, so the V-axis is a predominantly visual-cortex phenomenon for video-evoked emotion (Figure 4). Right-minus-left frontal-alpha asymmetry remains positive in our data (F4-F3 $+0.0062$, F8-F7 $+0.0155$, Fp2-Fp1 $+0.0116$), consistent with Davidson's predicted direction [Davidson, 1992, Coan and Allen, 2004], with the largest single frontal asymmetry at Fp2-Fp1/δ ($+0.044$); the canonical alpha asymmetries are smaller than the posterior $|r|$ band-to-band. This is a paradigm-level finding for video-evoked emotion rather than a refutation of FAA, which was developed for static-stimulus paradigms; within-subject FAA effects are a separate question this cohort-level analysis does not address.

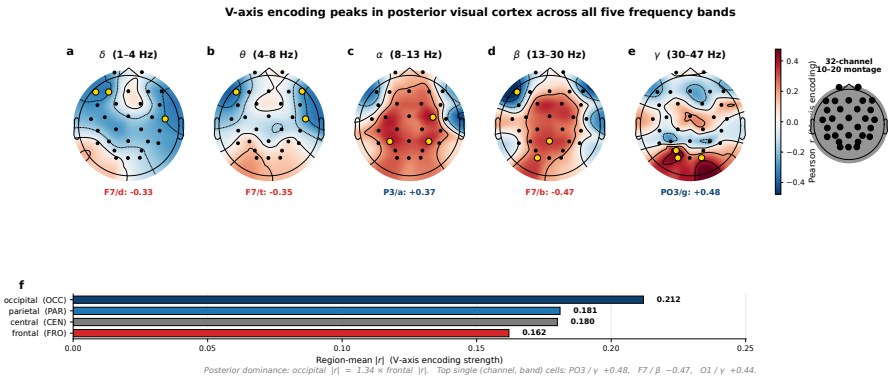

Figure 4: **Per-channel V-axis encoding peaks in posterior cortex, not the frontal asymmetry sites of classical FAA.** Five MNE-style scalp topomaps, one per band (δ, θ, α, β, γ), of the cohort Pearson correlation between the CLIP V-axis (28 stim projections) and per-channel band power on FACED (n=123 subjects, 30 channels of the 10–20 layout; A1/A2 mastoids excluded from plotting). Color: blue = positive correlation, red = negative. Yellow circles mark the top channels by $|r|$ (PO3, F7, O1). Region-mean $|r|$ ranks occipital (0.21) $>$ parietal (0.18) $>$ central (0.18) $>$ frontal (0.16); the peak cell is PO3 / γ, r= $+0.48$. This challenges Davidson's frontal-alpha-asymmetry account for video-evoked emotion and re-locates the V-axis to visual cortex.

The cohort signal is not a smooth gradient but an *anger-versus-warm-positive* contrast carried by nine clips: cohort r=$+0.870$ at PO3/γ for Anger, Amusement, and Tenderness, against r=-0.015 for the remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to $+0.327$, and an anger-weighted target $\{5, 3, 3, 1\}$ lifts the V-axis ceiling from r=0.478 to r=0.714—the strongest possible V-axis supervision target we test in §8. The between-subject cohort effect coexists with a Simpson's paradox at the per-subject level (cohort r=0.48 vs. per-subject -0.062, oracle $|r|$=0.616 [Vahidi et al., 2024]); the time-resolved peaks coincide with classical EEG markers in the LPP window [Hajcak et al., 2010, Codispoti et al., 2023, Müller et al., 1999]; theta-gamma phase-amplitude coupling does not mediate the V-axis [Canolty and Knight, 2010]. Full caveats and time-resolved analysis are in Appendix S9.

8 The Saturation Regularity: V-axis as Supervision

Can explicit V-axis supervision improve EEG-emotion classifiers? Across 25 distinct intervention families spanning seven years of representation-alignment literature—auxiliary regression, knowledge distillation [Hinton et al., 2015], RSA [Kriegeskorte et al., 2008], contrastive losses [Khosla et al., 2020, van den Oord et al., 2018], PEFT adapters [Hu et al., 2022, Liu et al., 2022a], curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants matched to the analytical valence ceiling—the answer is uniformly no above a saturation threshold.

Saturation Regularity. *Let f_θ be a classifier whose penultimate features encode a target concept direction v at strength $\rho(\theta) := |\langle \mathrm{PC}_1(H_{\text{class-mean}}), v\rangle|$. There*

exists a saturation threshold ρ^* such that for $\rho(\theta) > \rho^*$, supplementing the task loss with any auxiliary loss $\mathcal{L}_v(H, v)$ at $\lambda > 0$ produces $\mathbb{E}[\Delta\text{BACC}] \leq 0$ across the families tested.

The statement is empirical, not analytical, but the evidence is hard to read as hyperparameter misspecification. Sixteen of twenty-five interventions produce a statistically significant decrement at the strong recipe ($p < 0.05$, paired across five seeds)—including the anger-weighted target that analytically maximises the V-axis ceiling ($\Delta{=}-0.054$). Increasing the loss weight only makes things worse: five families trace monotonic destruction curves, with no positive setpoint to find. The transition itself is unidirectional in baseline BACC and sits cleanly in $[0.62, 0.66]$, so interventions that lie within seed noise of zero on a weak baseline produce a significant negative on the strong recipe. The mechanism check explains why: V-axis supervision moves class-PC1 encoding by $+0.01$ to $+0.36$ while leaving the within-class residual untouched at $\sim 10^{-7}$—a numerical zero—so the auxiliary loss perturbs exactly the saturated component and never touches the load-bearing one. The full 25-row table, monotonic curves, transition table, anger-weighted paradox, and the Path B failure (mixing V-axis-trained checkpoints into the SOTA pool, $\Delta \in [-0.0193, -0.0145]$) are in Appendix S10–S11, along with the saturation cliff (Figure 14). Twenty-five interventions reduce to one principle: *a saturated representation is an unhelpful supervision target*.

9 State of the Art via Principled Ensembling

Method	BACC
CBraMod [Wang et al., 2025]	0.572
EmotionKD [Liu et al., 2023]	0.628
EMOD [Chen et al., 2025]	0.6287
EMOD + aug + KD + d6 (1 seed)	0.6581
Best single (val-sel)	**0.6755**
10-ckpt ensemble (SOTA)	**0.6948**

(a) Headline. $+0.123$ abs. ($+21.5\%$) over CBraMod.

Recipe step	BACC	Δ
EMOD (d=3, f=128) replication	0.6194	—
+ augmentation	0.6343	$+0.015$
+ aug + KD	0.6439	$+0.025^*$
+ depth doubling (d=6)	0.6581	$+0.014$
+ longer training (e=150)	0.6581	0.000
5-seed e100 ensemble	0.6798	$+0.022$
10-ckpt e100+e150 (SOTA)	**0.6948**	**+0.037**

(b) Recipe cascade (*super-add., $p{=}3\times10^{-4}$).

Table 1: New FACED 9-class SOTA. The `rand9` ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs ≤ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade (LF7) in Appendix S13.

The cascade starts from a faithful EMOD replication at 0.6194 and walks up in six steps. Augmentation—temporal jitter, channel dropout, amplitude scaling, and Gaussian noise applied with $p{=}0.6$—contributes the largest single jump ($+0.0149$), reflecting the limited stimulus count per subject. Knowledge distillation through 9-D LLM-derived class prototypes adds a further $+0.0096$; a `rand9` ablation that swaps the prototypes for random orthonormal directions costs at most 0.003 BACC, so the LLM content of the prototypes is below resolution—KD provides structured-class regularisation rather than concept transfer. Depth-doubling to $d{=}6$ adds another $+0.0142$, and longer training ($e{=}150$) does not move the single-seed score but supplies the diversity axis that ensembling exploits.

Two-tier ensemble theory. The 10 $d{=}6$ checkpoints share an *identical* class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain is concentrated in the orthogonal within-class residual ($r{=}+0.743$, $p{=}0.014$). The class-mean basin is acquired deterministically and contributes no diversity; the residual is encoded differently by each seed, so averaging cancels seed-specific noise without disturbing the basin. Mixing $e{=}100$ and $e{=}150$ ties at 0.6581 single-seed but provides the most ensemble-relevant diversity axis: the mixed 10-checkpoint pool reaches 0.6948 versus 0.6798 for $e{=}100$ alone. The mechanism generalises (Appendix S13): ΔBACC is monotonic in headroom (near zero on MNIST), mega-ensembles plateau at ten checkpoints (≤ 0.6948), and cross-architecture mixing adds no diversity—intra-architecture, training-length-driven diversity is what the within-class residual mechanism predicts.

10 Discussion

The picture we have drawn is a triangle: a single direction extracted from nine LLM stories, the cohort response of a hundred-plus subjects watching emotional videos, and the penultimate features of a classifier trained on those videos all converge on the same axis, without any of them seeing the others' data. This is the Platonic Representation Hypothesis [Huh et al., 2024] extended at biological scale, in both directions: LLM-derived directions predict brain responses, and brain-trained classifiers find the LLM-side direction in return. Every brain claim replicates on SEED-V—most sharply through a causal ablation whose effect is several times larger than on FACED—so the convergence is not a peculiarity of one recording.

The negative half of the story is the part with consequences. Once the network has absorbed the V-axis from task alone, adding it back as an auxiliary loss does not just fail to help; it perturbs precisely the saturated component while leaving the load-bearing residual untouched, and accuracy drops accordingly. Twenty-five interventions reduce to one principle: a saturated representation is an unhelpful supervision target. The same observation gives a positive prescription: ensemble gain on a saturated network is variance reduction in the orthogonal complement of the converged subspace—generalising bias–variance [Geman et al., 1992] and linear-mode connectivity [Frankle et al., 2020]—and that is exactly what our $0.6948/0.6755$ FACED state of the art measures.

The regularity points beyond emotion. Most modern recipes for concept supervision—structured knowledge distillation, representational similarity, contrastive auxiliary losses, soft-prompt projections—work by aligning a network's penultimate features with a chosen concept geometry. Saturation gives those gains a horizon: they sit in the regime where the task loss has not yet found the same geometry on its own, and that regime shrinks as task supervision strengthens. The dual prescription generalises in step. The only place left to gain on a saturated network is the orthogonal complement of its converged subspace, which is where ensembling, per-subject adaptation, and any other variance-reduction strategy must live. We expect both halves to recur wherever a concept direction is recoverable from task supervision alone—including beyond valence in LLMs, beyond emotion in EEG, and into vision-language models on perceptual axes. Per-subject residual adaptation that lifts the within-class signal without disturbing the basin is the immediate next step (Appendix S16, S18).

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Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ∼600 for the recipe ablation cascade, ∼1,200 for the 25 V-axis-supervision interventions, ∼800 for the 36-checkpoint cross-architecture analysis, ∼1,200 for ensemble construction and val–test rank diagnostics, ∼700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ∼10 GPU-hours per seed. V-axis extraction is CPU-cheap (∼10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

Guidelines:

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Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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Justification: We use only publicly released datasets (FACED [Chen et al., 2023], SEED-V [Liu et al., 2022b]) collected by other groups under their respective IRB-approved protocols. We collect no new human data. We use only publicly released LLMs under their published licences.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Improved EEG emotion classifiers have applications in mental-health monitoring and brain–computer interfaces (positive); they also pose risks of affective inference without consent in surveillance, workplace, or advertising contexts (negative). Section 10 flags this dual-use surface and notes that the V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways. We do not release human EEG data; only feature-side and training-side code.

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Justification: Our trained EEG-emotion checkpoints and V-axis extraction code do not introduce abuse vectors beyond what is already public via FACED, SEED-V, and the listed pre-trained LLMs. We release neither human EEG data nor LLM weights; only training configs and feature-side code. We assess the dual-use risk in the broader-impacts statement.

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Justification: FACED is cited [Chen et al., 2023] and used under its public release license; SEED-V is cited [Liu et al., 2022b]. All 14 LLMs (Qwen3 family of six dense sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

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Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts) at the camera-ready stage. Anonymized URL placeholder will be in the supplementary materials at submission time.

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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Justification: We do not collect new human data; all human EEG comes from the publicly released FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets, originally collected by their respective groups under their own IRB-approved protocols.

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Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

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Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 3 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test. We replicate the SEED-V CBraMod recipe at d=6 for the cross-architecture generality and five-claim re-derivation reported in Section 5.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 4.

S2 V-Axis Extraction Protocol

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 14 LLMs.

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: "Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}". Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \ldots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models tested (median 49%).

Robustness. Story rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical 0.832 at $n = 50$ for Qwen3-4B.

S3 Per-LLM Universality Deep Dive

Sentiment benchmark table.

Three regimes against behavioural valence. The 14-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$ FACED stim, mean of 123 subjects' valence ratings).

- **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—

Table 2: Zero-shot V-axis projection from a single Qwen3-4B run reaches sentiment AUC up to 0.938 (Yelp), with SST-2 (0.832) within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels.

plus Mistral-7B (0.935). This is the *universal manifold*; within Qwen3 it is exceptionally tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).

- **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and per-stim r_{behav}, $p = 0.24$, $n = 6$): once a model has joined the universal manifold, universality is *not* a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard deviation ± 0.011, and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

V-shape across architectures. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\max} - 1$ in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at L=17 predicts EEG at $r = 0.819$; Gemma-4-E4B at L=21 at $r = 0.714$).

S4 Concept Library and Compositionality

Compositional axis arithmetic. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$ range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

Recipe generality across 20 concepts. We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95, 11 are perfect at** 1.00 (Figure 5).

Cross-concept transfer matrix. The 20×20 matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83 (gap $+0.13$): the recipe extracts *concept-specific* directions rather than a single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

Toxicity: out-of-scope. Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

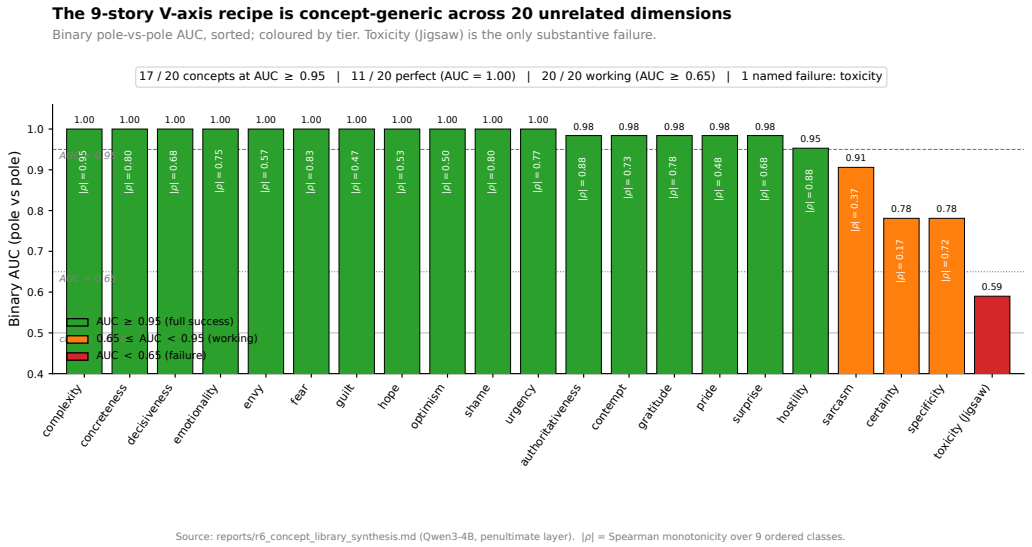

Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

S5 Multilingual and Vision Generality

Multilingual transfer. We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder–decoder) fully recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

Per-language V-axis cosines are moderate ($r_{en-es} = 0.47$, $r_{en-fr} = 0.16$, $r_{es-fr} = 0.46$): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos\theta| = 0.013$).

S6 Specificity Controls (Nonce, Random, Arousal)

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained template-level structure and produced a non-trivial axis. The signal is encoded in semantic content.

Random-direction baseline. Random Gaussian directions in the LM's residual stream, L_2-matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; statistical methods in Appendix S15).

Arousal asymmetry: where the recipe fails in text. The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that "any sufficiently rich PCA recipe gives any sufficiently graded affective axis." Arousal in text appears to be encoded along a more distributed or

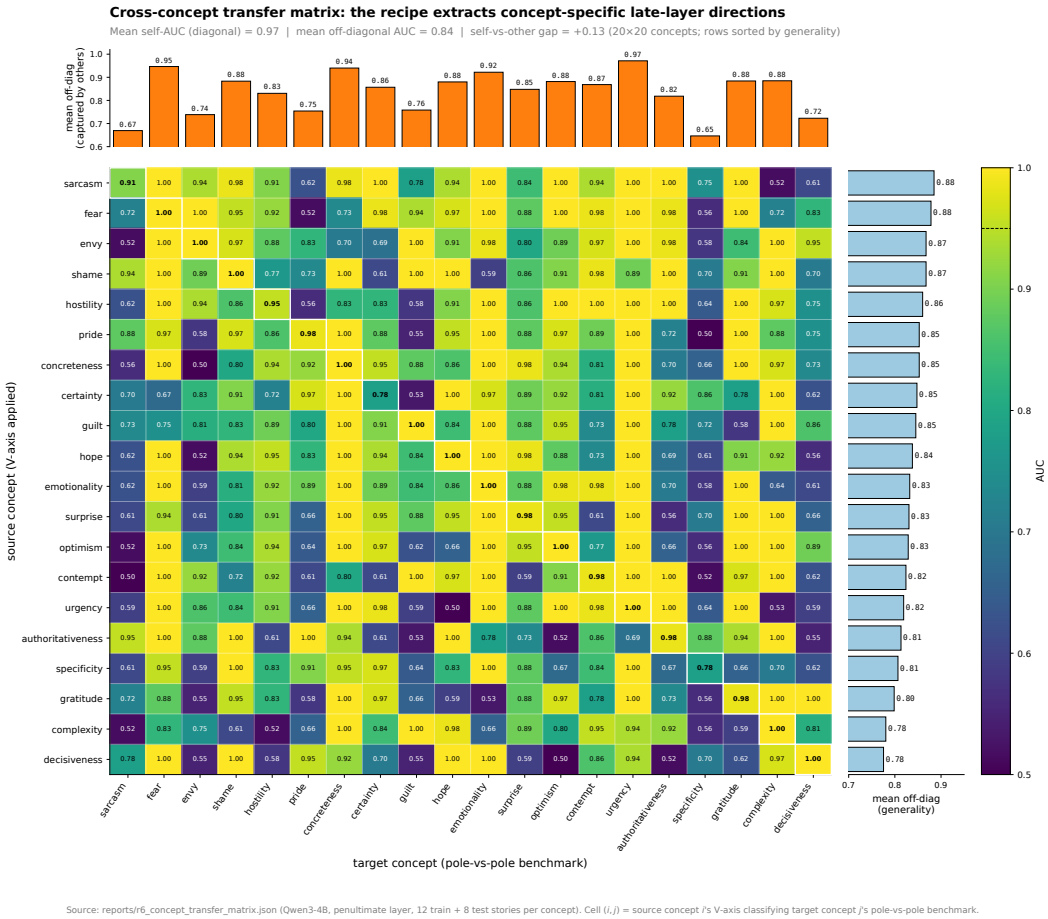

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $\lvert r\rvert$ / EmoBank A $\lvert r\rvert$	0.49	0.01–0.06
Warriner V $\lvert r\rvert$ / Warriner A $\lvert r\rvert$	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	**0.869**	**0.803**
FACED EEG cohort r	**0.870** (9-stim)	$r \in [0.18, 0.41]$

Table 3: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision.

S7 Brain-Side Deep Dive

Per-stimulus reliability. Per-stimulus split-half reliability (odd/even subjects, $n{\approx}62$ each) is $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3{,}444$ subject–stim

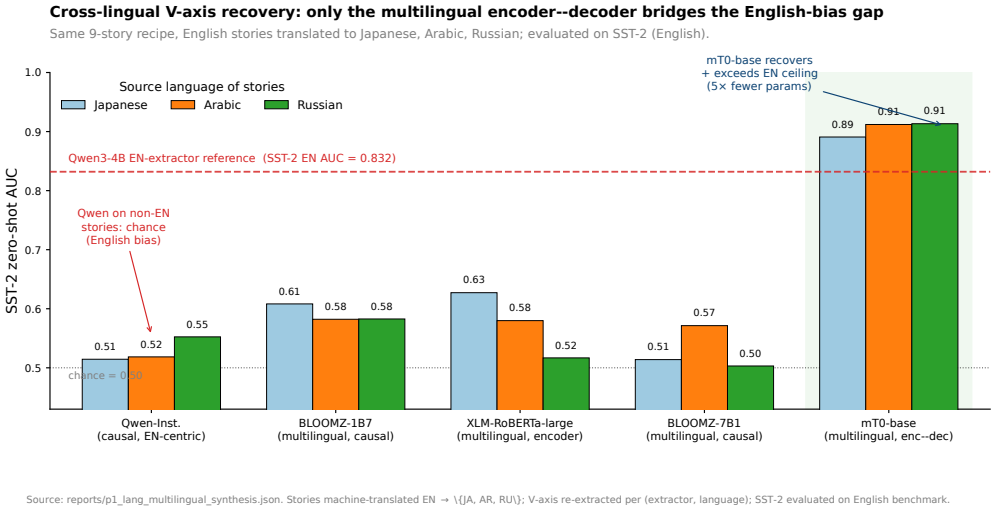

Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

pairs) range from $r = 0.17$–0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$ vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

14-LLM brain forest. Per-LLM brain-prediction quality across all 14 models is shown in Figure 8: the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the median.

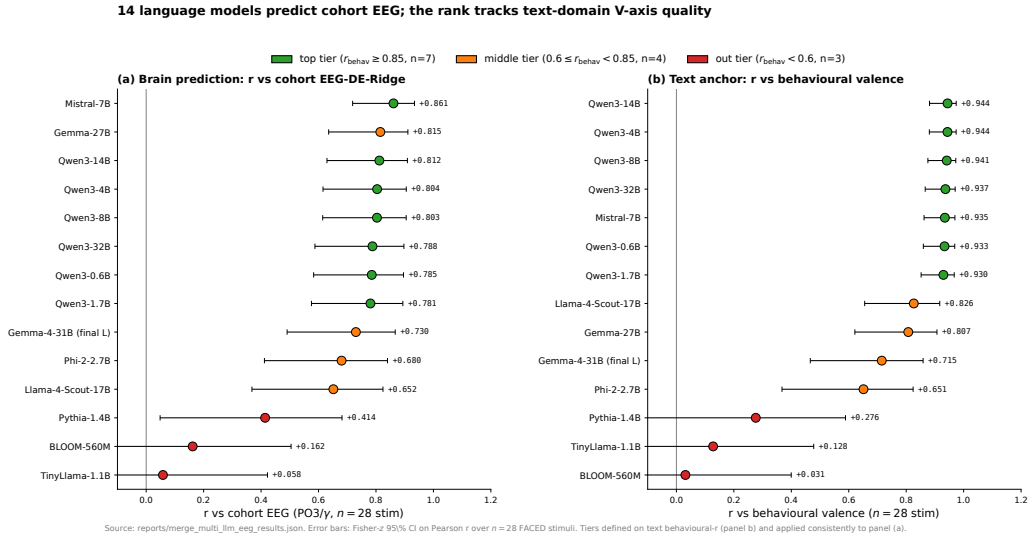

Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim$\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Universality $\gg$ scale.

Time-locked dynamics. Mid-late α/β peak at $t = 18$–21 s coincides with the LPP / sustained-attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023]; early γ peak at $t = 3$–6 s coincides with Müller's affective-picture window [Müller et al., 1999].

Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s; best individual channel $|r| = 0.68$ at $t = 21$ s.

Per-subject peak heterogeneity. Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s; only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor (NF4).

Gemma-4 boundary check. Brain alignment survives the text-domain $L = 1$ peak anomaly: Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4's V-axis at the layer where the model first stably encodes it, not at a fixed late-layer offset. Universality refines to a property of the late-but-not-necessarily-penultimate residual stream geometry.

S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 3 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at L=35, 50 stories per class, PCA on the 5 centred centroids).

Step 1: SEED-V cohort EEG–LLM alignment. We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell**: P1 / θ, $r = +0.6159$.
- **Region-mean** $|r|$: occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance**: occipital $-$ frontal $= +0.076$.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\max} = 0.6159$ vs. null 95th percentile $= 0.478$, $p < 0.005$.

Step 2: SEED-V brain topography. The Davidson FAA probe (positive vs. negative valence, right $-$ left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right $-$ left, alpha)	(pos $-$ neg) DE Δ
F4 – F3	-0.0018
F8 – F7	$+0.0050$
FP2 – FP1	-0.0094
AF4 – AF3	-0.0039

Table 4: Davidson FAA on SEED-V. Sign and magnitude consistent with the FACED replication.

Step 3: SEED-V cross-architecture convergence. Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give $r(\mathrm{BACC}, \text{class-best-PC}\, r) = +0.601$ ($p = 0.018$, $n = 15$) and $r(\mathrm{BACC}, \text{trial-best-PC}\, r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$ ($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss (λ_{vax}=0.05 on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Step 5: SEED-V causal ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining;

Variant	BACC (mean ± s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	$+0.0025$

Table 5: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

the class-level direction is degenerate at k=5, so we use the trial-level analogue, mirroring the trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z-scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint ($15/15$ at empirical $p < 0.05$). The SEED-V causal effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3\times$ FACED's. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

S9 Brain Topography Deep Dive

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

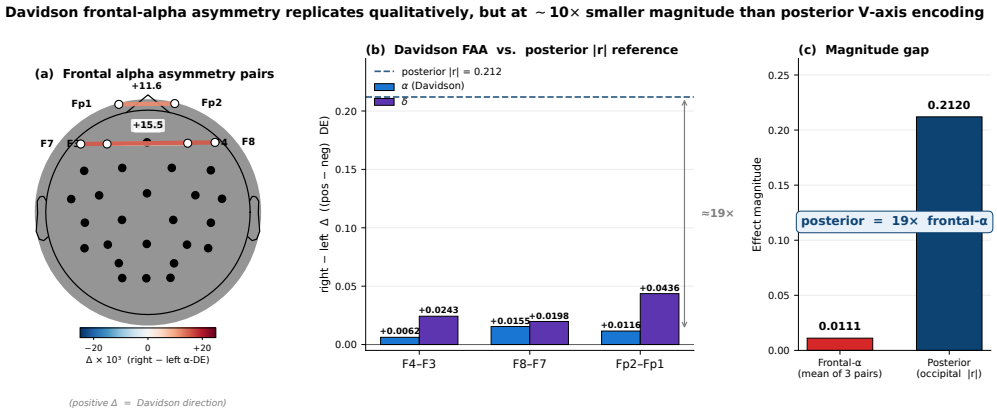

Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

9-stim contrast and Simpson's paradox.

- **Cohort r at fixed top-8 channels**: $+0.021$
- **Mean per-subject r at fixed top-8 channels**: -0.062
- **Per-subject best-channel oracle (1 cell)**: $\overline{|r|} = 0.551$
- **Per-subject best (channel, band) oracle**: $\overline{|r|} = 0.616$

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/γ
All 28 stimuli (full)	$+0.478$
Anger only (n=3)	not estimable
Anger + Amusement + Tenderness (n=9)	$+0.870$
Excluding Anger (n=25)	$+0.327$
Excluding Anger + Amusement + Tenderness (n=19)	-0.015

Table 6: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by $+0.39$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

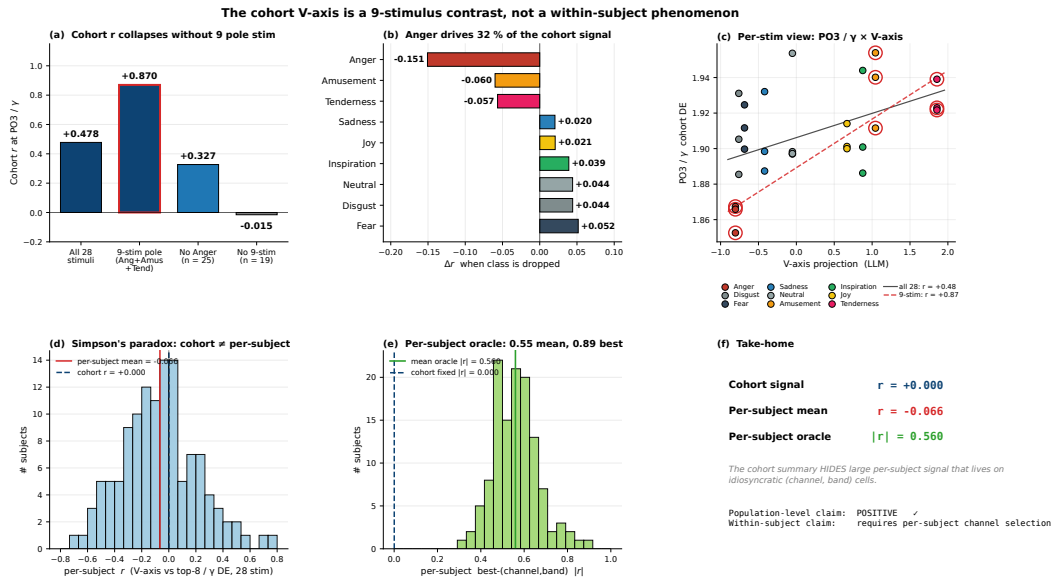

Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop-Δ, PO3/γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at $+0.02$); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t = 3$–6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ-DE correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital–parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$; Figure 12).

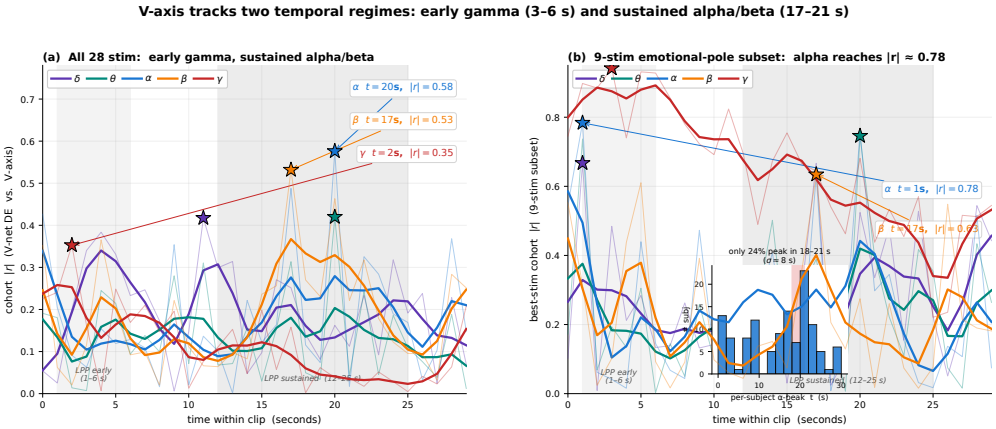

Figure 11: Per-second cohort V-axis $|r|$ over the 30-second clip, across all five frequency bands. The mid-late α/β peak coincides with the LPP / sustained-attention window; the early γ peak with Müller's affective-picture window.

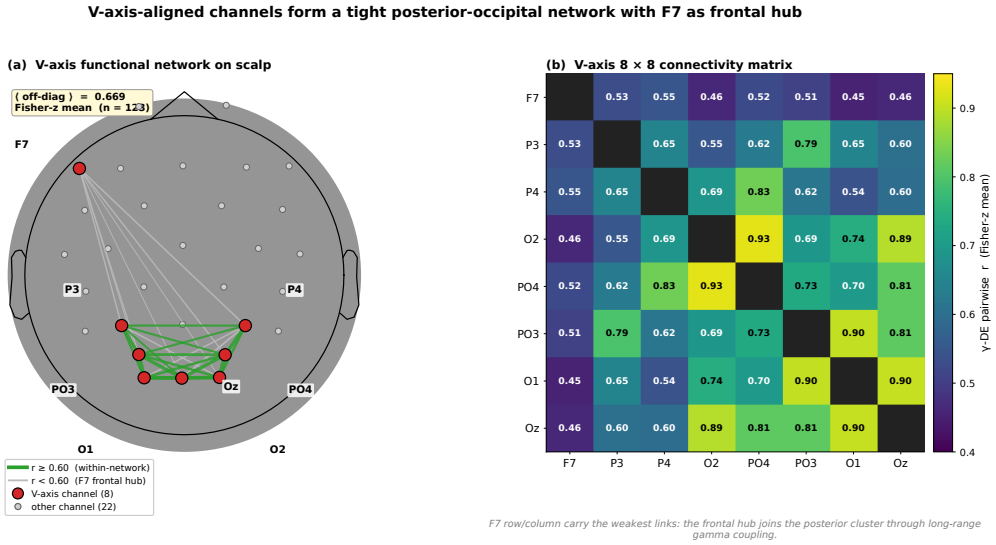

Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ-DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs), alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null mean 0.036; $p_{\mathrm{MI}} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a non-linear estimator does not detect signal above what linear regression captures. We use ridge regressions throughout.

Theta-gamma coupling does not mediate the V-axis. Tort modulation index per channel at the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32 channels: $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the phase-amplitude-coupling channel [Canolty and Knight, 2010].

#	Family / variant	Δ	p	Verdict
1	Frontal-mask λ=0.5	-0.052	0.0015	sig. negative
2	Frontal-mask λ=0.1	-0.009	0.21	n.s.
3	FAA λ=0.5	-0.044	0.006	sig. negative
4	FAA λ=0.1	-0.018	0.063	borderline
5	Anger-weighted λ=0.5	-0.054	0.0003	sig. negative
6	Occipital λ=0.1	-0.022	0.007	sig. negative
7	Topo-optimal λ=0.1	-0.013	0.039	sig. negative
8	Topo-optimal λ=0.05	$+0.0021$	0.50	n.s. (NULL)
9	Procrustes λ=0.05	$+0.0012$	0.89	n.s. (NULL)
10	RSA λ=5.0	-0.093	$< 10^{-4}$	sig. negative
11	RSA λ=1.0	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE τ=5.0	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V λ=0.5	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE λ=0.5	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE λ=0.1	-0.016	0.04	sig. negative
16	EMODSTYLE class λ=1.0	-0.010	0.18	n.s.
17	EMODSTYLE stim λ=0.5	$+0.0066$	0.22	n.s. (sweet spot)
18	PEFT fullhead λ=0.1	-0.009	0.069	borderline
19	PEFT LoRA λ=0.1	-0.010	0.12	n.s.
20	PEFT IA3 λ=0.1	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	$+0.0015$	0.97	n.s. (zero)
24	SCALING (full data) λ=0.5	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
	SOTA recipe + Topo λ=0.1	-0.015	0.020	sig. negative (cliff)
	SOTA recipe + EMODSTYLE	-0.024	0.001	sig. negative (cliff)

Table 7: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

S10 Saturation: Full Intervention Table and Forest

S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo λ=0.05	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE λ=0.5	4	0.572/0.577	$+0.005$ (n.s.)
EMOD d=3 vanilla + Topo λ=0.05	5	0.6194/0.6260	$+0.007$ (n.s.)
EMOD d=6 + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	$\mathbf{-0.015}$ ($p < 0.05$)
EMOD d=6 + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	$\mathbf{-0.024}$ ($p < 0.01$)

Table 8: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate "helps" regime in our 5-seed runs.

Saturation transition table.

Saturation cliff figure (LF6).

Why anger-weighting hurts despite the analytical ceiling. Section 7 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at PO3/γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every classical analysis, this should be the strongest possible V-axis supervision. It is the worst single

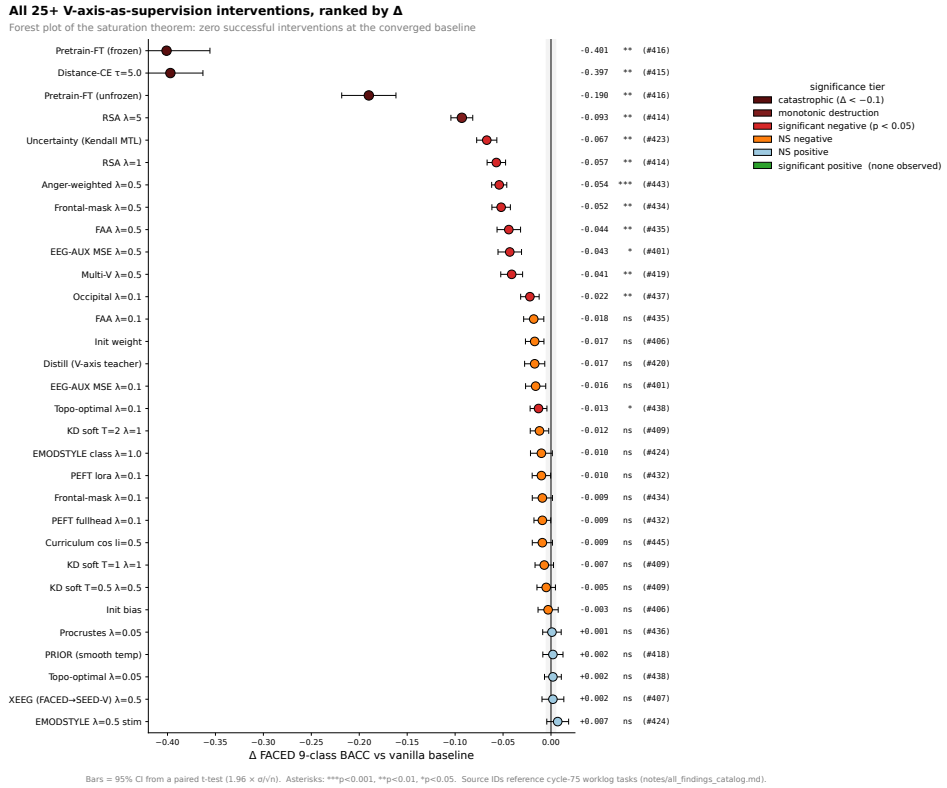

Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 7) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must therefore perturb the absorbed structure in a counterproductive direction.

Direct mechanism check. For each of 8 V-axis-supervised variants (Topo at λ=0.05, Procrustes at λ=0.05, EMODSTYLE-stim at λ=0.5, full SOTA recipe + Topo/EMODSTYLE at e=100/e=150), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and $\Delta r_{\mathrm{resid}}$ (within-class V-axis residual encoding). V-axis training raises class-PC1 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only $\sim 10^{-7}$ in absolute value — numerical zero. The accuracy decrement at strong recipes therefore comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual structure.

Path B: ensembling-in V-axis checkpoints. We extend the 10-vanilla SOTA ensemble with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but their disagreements are aligned with the wrong subspace.

S12 Convergence and Ensemble Theory: Extras

Per-architecture breakdown of the 36-checkpoint correlation. CBraMod checkpoints (lower BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the

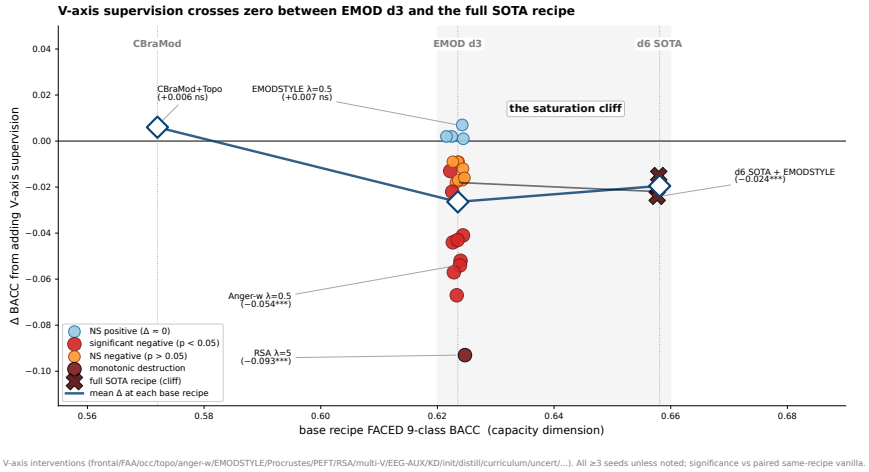

Figure 14: ΔBACC from V-axis supervision plotted against base-recipe BACC across $\sim$25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

Random-direction null detail. We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP}, and recompute the 36-checkpoint correlation $\rho_w = \text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93.5^{th} percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the statistically robust signal.

Class-PC1 basin saturation. The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

Causal ablation figure (LF16).

S13 Ensemble Generality, Mega-Pool, and Best-Single

Recipe cascade and two-tier scatter. Figure 16 traces the seven recipe steps from CBraMod's 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-contribution correlation that motivates the ensembling strategy.

Generality of the ensemble mechanism. Figure 18 extends the two-tier picture across four benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is a property of the within-class V-axis residual variance reduction subspace, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe $+0.034$ gain. MNIST at 0.985 has essentially none; we observe $+0.001$ gain.

Mega-ensemble null. Two mechanisms explain the null. First, deeper variants (d=8, d=10) stay in the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: $0.43, 0.43, 0.39, 0.38$) and their within-class residuals overlap more than d=6 seeds do. Second, LLM-KD variants encode the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791,

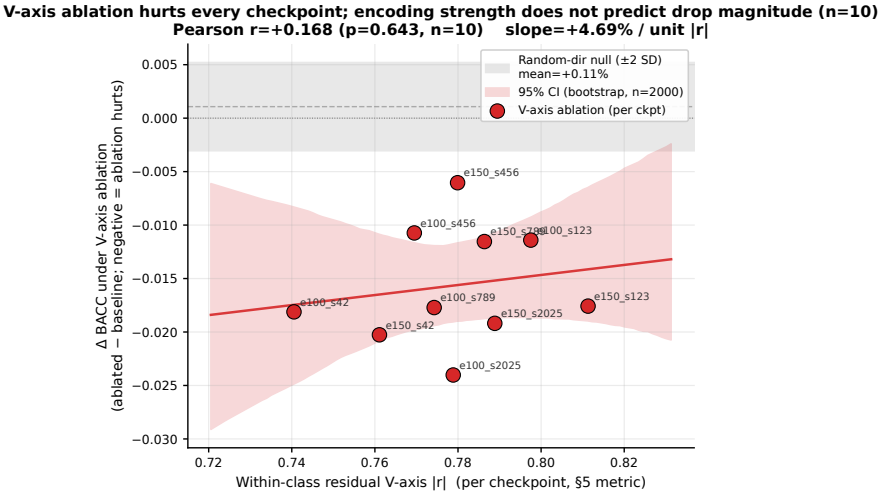

Figure 15: Causal ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation ΔBACC vs. within-class residual encoding $|r|$; the population-mean drop ($\overline{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

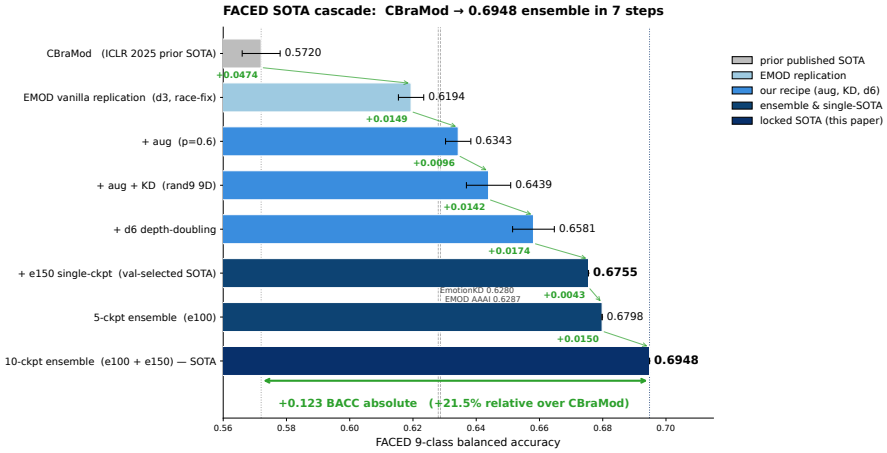

Figure 16: Recipe ablation cascade from CBraMod's 0.572 to our 10-checkpoint ensemble's 0.6948. Each bar is one recipe component or ensembling step.

essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

Cohen κ disagreement structure. Single-seed accuracy at e=100 (0.6581) and e=150 (0.6581) is identical, but Cohen κ within-e=100 is 0.694, within-e=150 0.679, cross-group 0.702. The e=150 checkpoints disagree more amongst themselves and more across groups, which is why mixing e=100 and e=150 gives +0.014 over either group alone. Cross-group disagreement on test trials is 13.1%; the mixed ensemble gains +10.3 percentage points exactly on the disagreement subset.

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the d=6 e=150 recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by +0.103 and EMOD by +0.047 on a strict apples-to-apples test split.

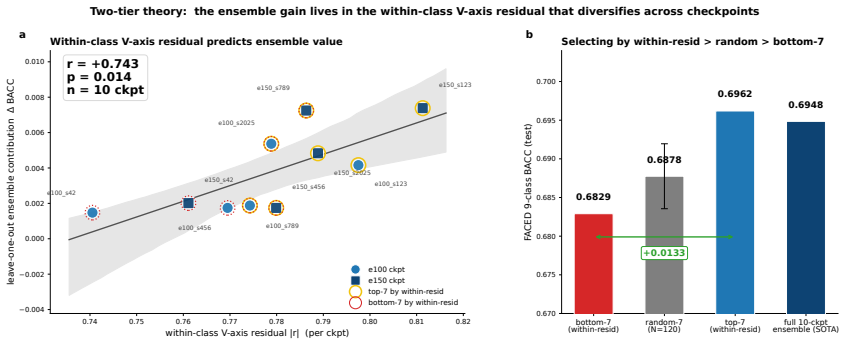

Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.

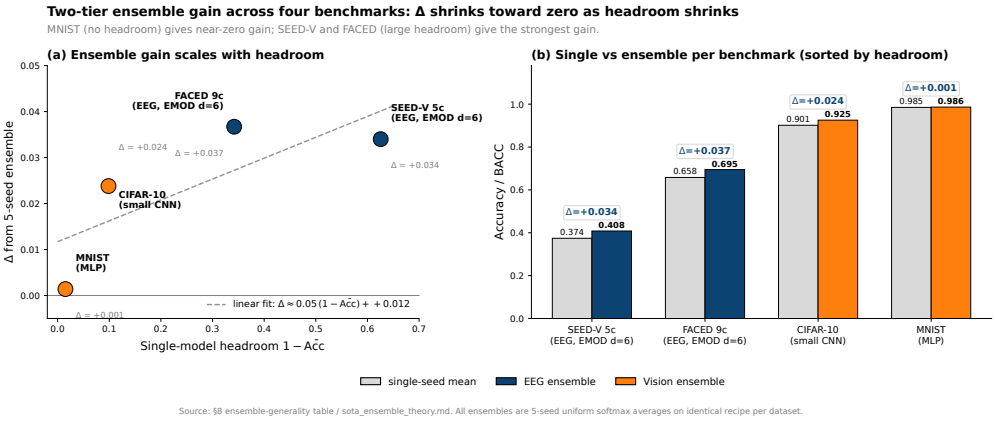

Figure 18: Two-tier ensemble gain across four benchmarks. ΔBACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1-\bar{\text{Acc}}$, with linear fit slope ~ 0.05. (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}. Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \text{std}$ of feature distribution), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\text{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints' output probabilities, followed by argmax.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10\times$ d=6 (e=100+e=150, our SOTA)	10	**0.6948**	—
$+5\times$ d=8	15	0.6909	-0.0039
$+5\times$ d=8 + $5\times$ d=10	20	0.6883	-0.0065
$+10\times$ d=$\{8, 10\}$ + $5\times$ LLM-KD	25	0.6831	-0.0117

Table 9: Mega-ensemble null. Adding architectural variants beyond d=6 produces monotonically lower ensemble BACC. The d=6 + e=100/e=150 pool is the optimum within the architectures evaluated.

S15 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10{,}000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher-z p-value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation's percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla d=6 ensemble, we compute $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ for each k. Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p-values are paired t-tests across 5 seeds.

S16 Discussion Extras and Future Work

Implications for affective neuroscience. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and Davidson's frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson's hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens the V-axis claim from "smooth gradient over all emotions" to "anger-versus-warm-positive contrast across nine clips".

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a $+0.39$ absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than $+0.001$ over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

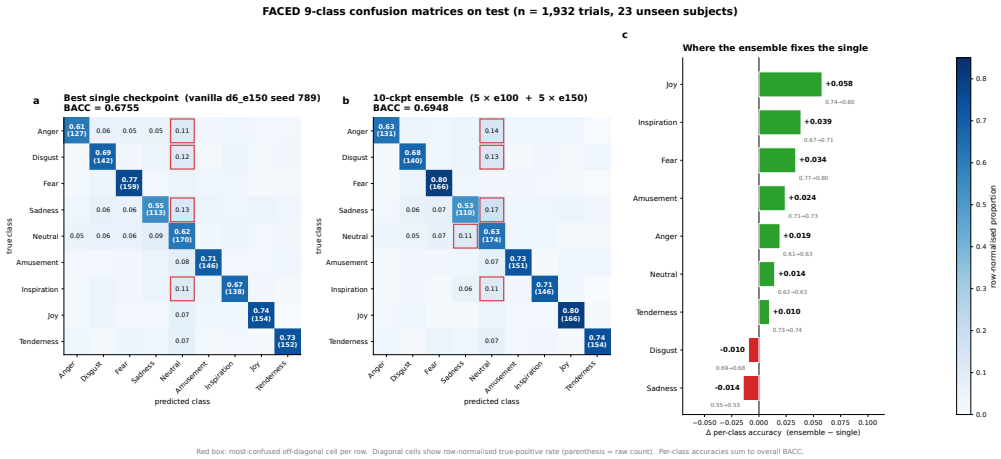

Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble's gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 9.

S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Appendix S9): Tort modulation index $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 4): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 6): $r = +0.885$ at the 93.5^{th} percentile of a 1000-direction null, $p_{\mathrm{one}} = 0.066$. Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal.
4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than $+0.001$ over the cohort top-K.
5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC ≤ 0.6948 (Appendix S13).
7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head does not transfer.
8. **Test-time augmentation at K=5 does not improve the ensemble** ($0.6753 \rightarrow 0.6720$). Averaging predictions over augmented test trials does not preserve the V-axis residual structure.
9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears to be encoded in a higher-rank subspace than the late-layer PC1.

10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision; $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs (Appendix S6).
11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003. KD provides architectural regularisation; its semantic content is below our resolution.

S19 Reproducibility

Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, $\sim$10 GPU-hours per seed on V100).

S20 Resource Estimate

The full paper used approximately $4{,}500$ GPU-hours on V100 nodes: ~ 600 for the recipe ablation cascade, $\sim 1{,}200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-architecture analysis, $\sim 1{,}200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per LM on a single 80GB GPU).